STRAVA FITNESS DATA ANALYSIS

SQL Business Insights Report

Project: Strava Fitness App – Case Study

This report presents the business insights generated from the cleaned Strava activity, hourly activity and sleep datasets. The findings below are based on the SQL analysis performed in MySQL Workbench.

Executive Summary

The SQL analysis shows clear differences in activity behaviour, sedentary time, sleep patterns and calorie expenditure. Higher activity levels are associated with substantially higher average daily steps, calories burned and very-active minutes. The analysis also identifies the most active users, the hours with the highest recorded step activity, and differences between sleep-duration groups and weekday/weekend behaviour.

Business Insight 1 – User Activity Segmentation

Activity Segment	Users	Average Steps
Low Activity	8	2,936
Moderate Activity	18	7,624
High Activity	7	12,488

Interpretation: Higher activity segments show a strong increase in average daily steps, with high-activity users averaging more than four times the steps of low-activity users.

Business Insight 2 – Activity and Sedentary Behaviour

Activity Segment	Users	Avg. Sedentary Minutes
Low Activity	8	1,174
Moderate Activity	18	931
High Activity	7	974

Interpretation: Low-activity users have the highest average sedentary time. Moderate-activity users record the lowest sedentary time among the three groups.

Business Insight 3 – Steps, Calories and Activity Level

Activity Segment	Users	Avg. Steps	Avg. Calories
Low Activity	8	2,936	2,014
Moderate Activity	18	7,624	2,298
High Activity	7	12,488	2,549

Interpretation: Average calorie expenditure rises with activity level, alongside a strong increase in daily steps.

Business Insight 4 – Sleep Duration and Activity

Sleep Segment	Users	Avg. Sleep Minutes	Avg. Daily Steps
Less than 6 Hours	8	230	7,693

Sleep Segment	Users	Avg. Sleep Minutes	Avg. Daily Steps
6–8 Hours	14	433	7,693
More than 8 Hours	2	579	4,073

Interpretation: The 6–8 hour group contains the largest number of users. The more-than-8-hour group has the lowest average daily steps, although it contains only two users.

Business Insight 5 – Sleep Efficiency

Sleep Efficiency Segment	Users	Avg. Efficiency	Avg. Sleep Minutes
Low Efficiency	2	65.6	473
High Efficiency	22	93.6	369

Interpretation: Most users fall into the high-efficiency sleep segment. High-efficiency users show considerably higher average sleep efficiency.

Business Insight 6 – Activity Intensity Distribution

Intensity Segment	Records	Very-Active Min.	Fairly-Active Min.	Lightly-Active Min.
Low Very-Active	536	1.0	7.4	178.0
Moderate Very-Active	151	18.2	21.0	230.0
High Very-Active	253	65.6	22.2	201.9

Interpretation: Very-active minutes increase sharply across the intensity groups, while lightly-active minutes are highest in the moderate very-active segment.

Business Insight 7 – Activity Intensity and Calories

Intensity Segment	Records	Avg. Sedentary Min.	Avg. Calories
Low Very-Active	536	1,034.0	2,029
Moderate Very-Active	151	948.3	2,358
High Very-Active	253	926.2	2,853

Interpretation: Higher-intensity activity is associated with higher average calorie expenditure and lower average sedentary minutes.

Business Insight 8 – Weekday vs Weekend Activity

Day Type	Activity Records	Avg. Steps	Avg. Calories	Avg. Sedentary Min.
Weekday	695	7,669	2,302	996
Weekend	245	7,551	2,310	977

Interpretation: Average steps are slightly higher on weekdays, while average calories are marginally higher and sedentary minutes slightly lower on weekends.

Business Insight 9 – Top 10 Most Active Users

Rank	User ID	Activity Days	Avg. Daily Steps	Avg. Calories	Avg. Very-Active Min.
1	8877689391	31	16,040	3,420	66.1
2	8053475328	31	14,763	2,946	85.2
3	1503960366	31	12,117	1,816	38.7
4	2022484408	31	11,371	2,510	36.3
5	7007744171	26	11,323	2,544	31.0
6	3977333714	30	10,985	1,514	18.9
7	4388161847	31	10,814	3,094	23.2
8	6962181067	31	9,795	1,982	22.8
9	2347167796	18	9,520	2,043	13.5
10	7086361926	31	9,372	2,566	42.6

Interpretation: The most active users consistently record high daily step counts. Several top users have complete 31-day activity coverage.

Business Insight 10 – Activity Coverage of Top Users

Metric	Observation
Maximum activity days	31 days
Minimum among top 10	18 days
Dominant pattern	Most top users have 30–31 active days

Interpretation: The top-user analysis indicates that sustained participation is a major characteristic of the most active users.

Business Insight 11 – Most Active Hours by Steps

Hour	Activity Records	Total Steps	Avg. Steps/Record
18:00	906	542,848	599
19:00	906	528,552	583
12:00	922	505,848	549
17:00	906	498,511	550
14:00	921	497,813	541
13:00	921	495,220	538
16:00	907	450,639	497
10:00	929	447,467	482
11:00	927	423,534	457
09:00	931	403,404	433

Interpretation: The 18:00 hour records the highest total steps among the top ten hours, followed by 19:00. This suggests a strong concentration of recorded activity in the late afternoon and evening.

Business Insight 12 – Activity Intensity and Sedentary Time

Intensity Segment	Records	Very-Active Min.	Fairly-Active Min.	Lightly-Active Min.	Sedentary Min.
Low Very-Active	536	1.0	7.4	178.0	1,034.0
Moderate Very-Active	151	18.2	21.0	230.0	948.3
High Very-Active	253	65.6	22.2	201.9	926.2

Interpretation: The high very-active segment combines the highest very-active minutes with the lowest average sedentary minutes.

Business Insight 13 – Sleep Duration and Daily Activity

Sleep Segment	Users	Avg. Steps	Avg. Calories
Less than 6 Hours	8	7,693	2,261
6–8 Hours	14	7,693	2,411
More than 8 Hours	2	4,073	1,557

Interpretation: The 6–8 hour sleep group has the highest average calorie expenditure, while the more-than-8-hour group has lower average steps and calories.

Business Insight 14 – Weekday vs Weekend Detailed Comparison

Day Type	Records	Avg. Steps	Avg. Calories	Avg. Very-Active Min.	Avg. Sedentary Min.
Weekday	695	7,669	2,302	21.2	996
Weekend	245	7,551	2,310	21.0	977

Interpretation: Weekday and weekend activity levels are broadly similar. Weekdays show slightly more steps and very-active minutes, while weekends show slightly higher calories and lower sedentary time.

Business Insight 15 – Activity Level Comparison

Activity Level	Records	Avg. Daily Steps	Avg. Daily Calories	Avg. Very-Active Minutes
Low Activity	303	2,128	1,807	2.3
Moderate Activity	334	7,466	2,355	13.4
High Activity	303	13,337	2,744	48.5

Interpretation: The relationship between activity level and performance is pronounced: high-activity records average over six times the steps of low-activity records and substantially higher calorie expenditure.

Key Business Takeaways

• Higher activity levels are strongly associated with higher daily steps and calorie expenditure.

• Low-activity users record the highest sedentary time in the activity-segment analysis.

• Most users in the sleep-efficiency analysis fall into the high-efficiency category.

• The 6–8 hour sleep group represents the largest sleep-duration segment.

• Late afternoon and evening hours, particularly 18:00 and 19:00, show the highest total recorded steps.

• Weekday and weekend behaviour is relatively similar, with only small differences in average steps, calories and sedentary time.

• Consistent activity coverage is common among the most active users.

Note: This report reflects the SQL outputs and business-insight analysis performed in MySQL Workbench. It is intended as the PDF companion to the SQL analysis file for project submission.